Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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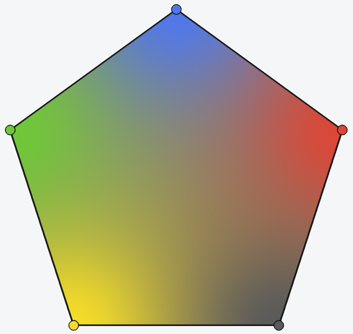
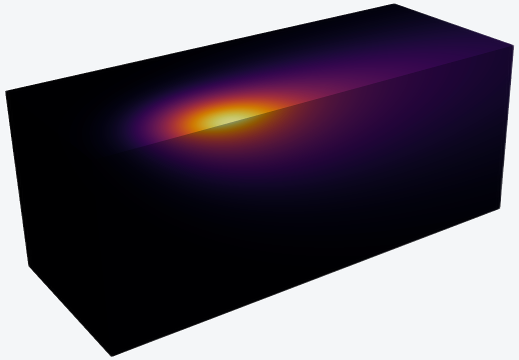
Core Thesis

Composition



- A 5-element affine projection represents the accessible composition simplex.
- Interior coloring encodes local element content across the design space.
- Bayesian updates narrow the search to feasible LPBF compositions.

Solidification Condition



- Real ‘eagar-tsai‘ VTI render with the symmetry-plane melt-pool slice.
- Process inputs set the thermal gradient, cooling rate, and pool geometry.

Solidified Microstructure

- Microstructure emerges from the coupling of chemistry and thermal history.
- Predicted G/R conditions map to expected grain morphology and phase selection.
- The workflow targets rapid microstructure prediction from composition to final structure.

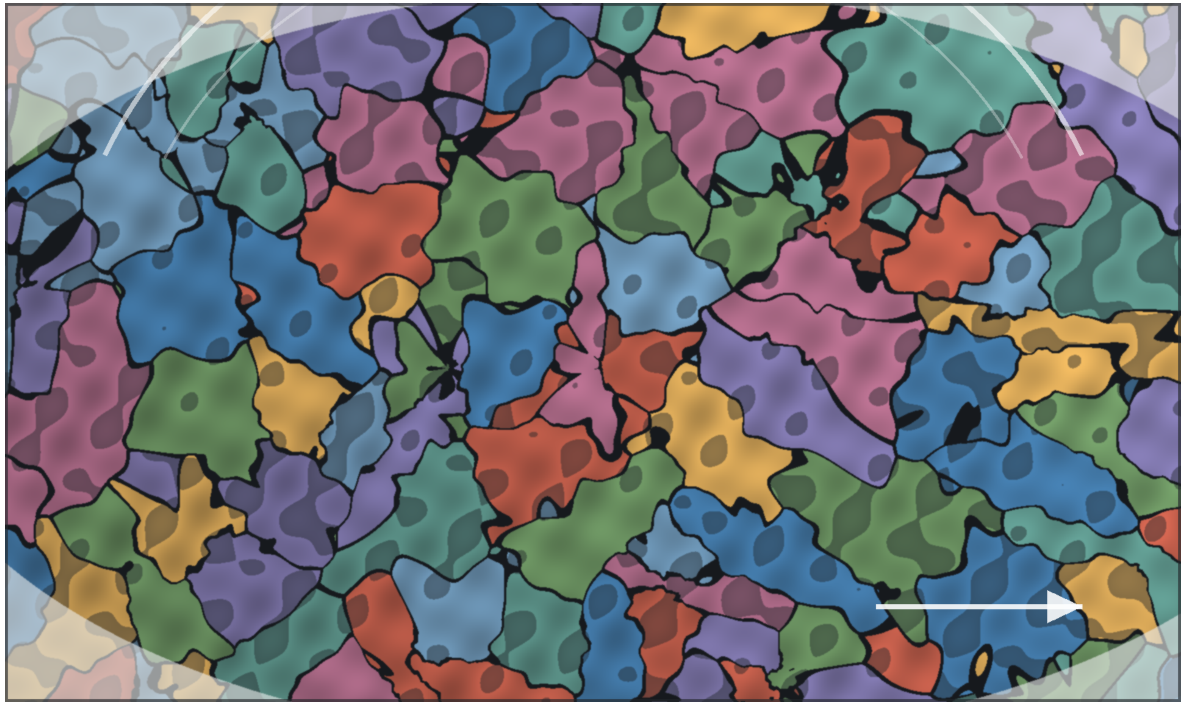
Motivation and Constraints

The design target is microstructure

- LPBF alloy design couples composition, laser power, scan speed, and cooling history.
- Grain morphology and phase selection respond to local thermal gradients, G, and solidification rates, R.
- Screening must move quickly from process inputs to a microstructure-relevant prediction.

Constraint

Eagar–Tsai is fast enough for high-throughput screening, but its simplified thermal physics must be corrected before it can be trusted for microstructure decisions.

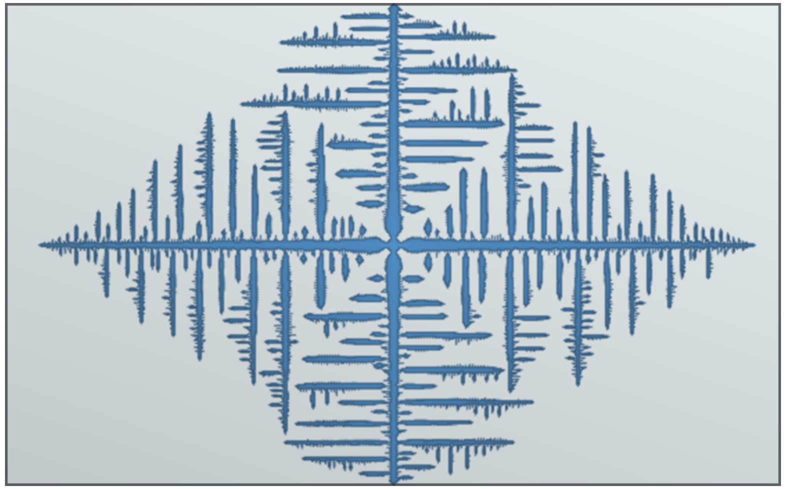


Illustrative phase-field style proxy: the model must connect melt-pool thermal history to grain morphology and phase contrast.

Key limits: temperature-invariant properties, analytical melt-pool assumptions, and lower fidelity than FEM.

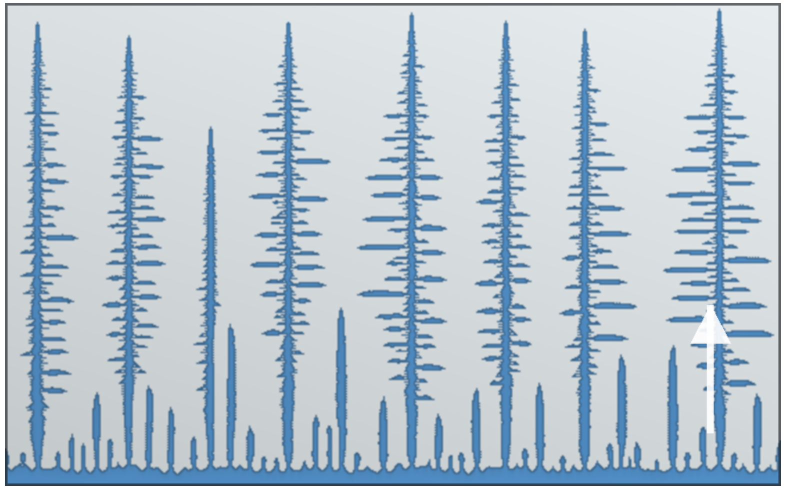
Cellular Automata Solidification Modes

Dendritic



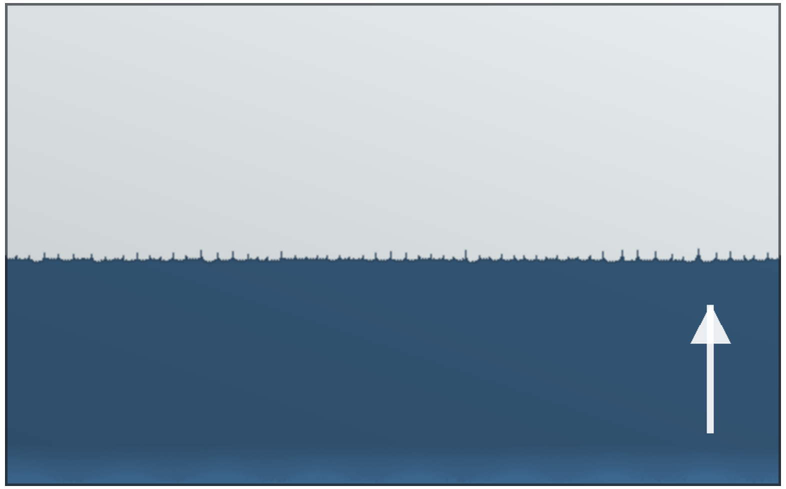
Branching growth when interface perturbations are amplified by undercooling and solute rejection.

Columnar



Competing primary trunks aligned with heat flow select λ_1; sidebranching sets λ_2.

Planar



A stable solid–liquid front advances when the thermal gradient suppresses morphological instability.

Use in this workflow

Eagar–Tsai supplies fast local G and R fields; Bayesian correction should make those fields reliable enough to classify morphology regime.

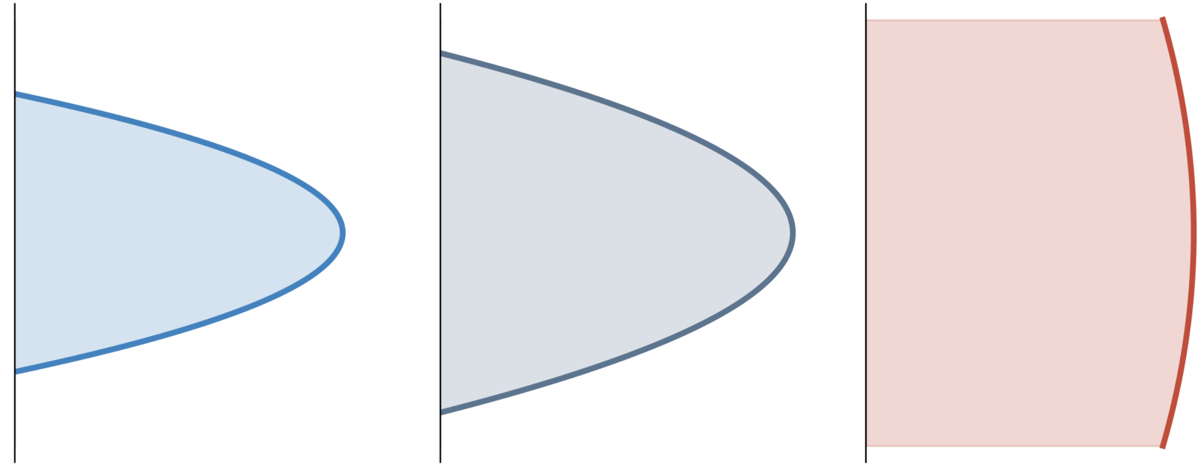
KGT Solidification Theory

Kurz–Giovanola–Trivedi framework

- KGT links dendrite-tip operating state to local thermal gradient, G, and interface growth rate, R.
- The dendrite tip is approximated locally as a paraboloid with radius of curvature ρ.
- Tip selection then relates ρ to R and alloy transport and capillarity properties.

Why it matters here

ET supplies local G and R along the solidification front. KGT uses those fields, together with material properties, to connect melt-pool conditions to dendrite-tip size and morphology selection.



Tip relations

$$z = \frac{x^2 + y^2}{2\rho}$$

$$Pe = \frac{\rho R}{2D_l}$$

$$\sigma^* = \frac{2D_l d_0}{\rho^2 R}$$

$$\rho = \left(\frac{2D_l d_0}{\sigma^* R}\right)^{1/2}$$

$$d_0 \approx \frac{\Gamma}{|m|\, C_0(1-k)/k}$$

- D_l: liquid diffusivity, Γ: Gibbs–Thomson coefficient, m: liquidus slope.
- C_0: nominal composition, k: partition coefficient, σ^*: tip-selection parameter.
- Higher R gives a smaller selected tip radius ρ for fixed material properties.

In this workflow, G determines the imposed thermal field and morphology stability, while R enters directly into the selected paraboloid tip radius.

Eagar–Tsai Model Foundation

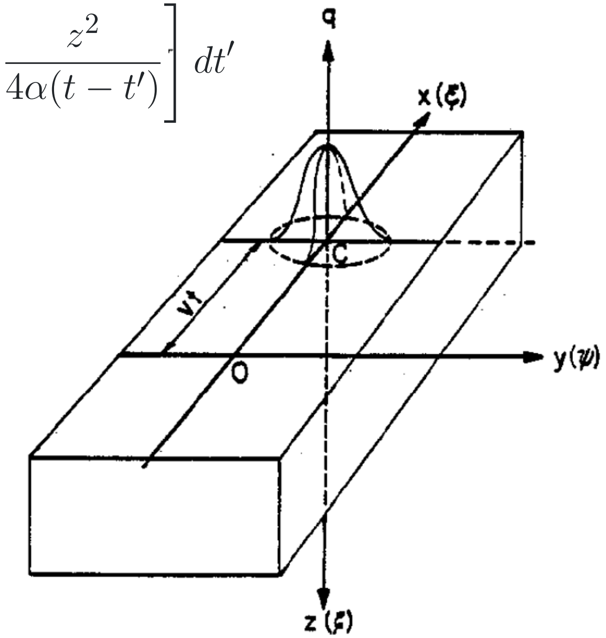
Original Eagar–Tsai Paper

Transient Gaussian moving heat source

$$T - T_0 = \frac{1}{2}\int_0^t dT_i$$

$$= \frac{q}{\pi\rho C_p\sqrt{4\pi\alpha}}\int_0^t \frac{(t-t')^{-1/2}}{2\alpha(t-t')+\sigma^2}$$

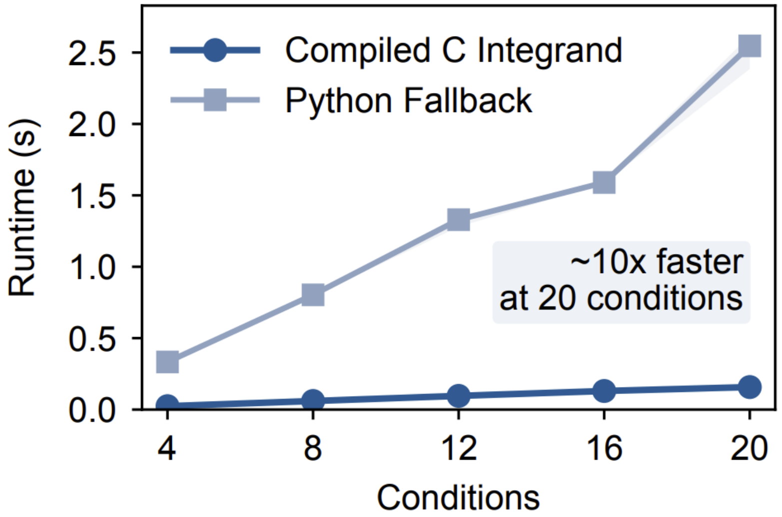
$$\times \exp\left[-\frac{(x-vt')^2+y^2}{4\alpha(t-t')+2\sigma^2}-\frac{z^2}{4\alpha(t-t')}\right]dt'$$



Eagar, T. W., & Tsai, N. S. (1983). Temperature fields produced by traveling distributed heat sources. *Welding Journal*, *62*(12), 346–355.

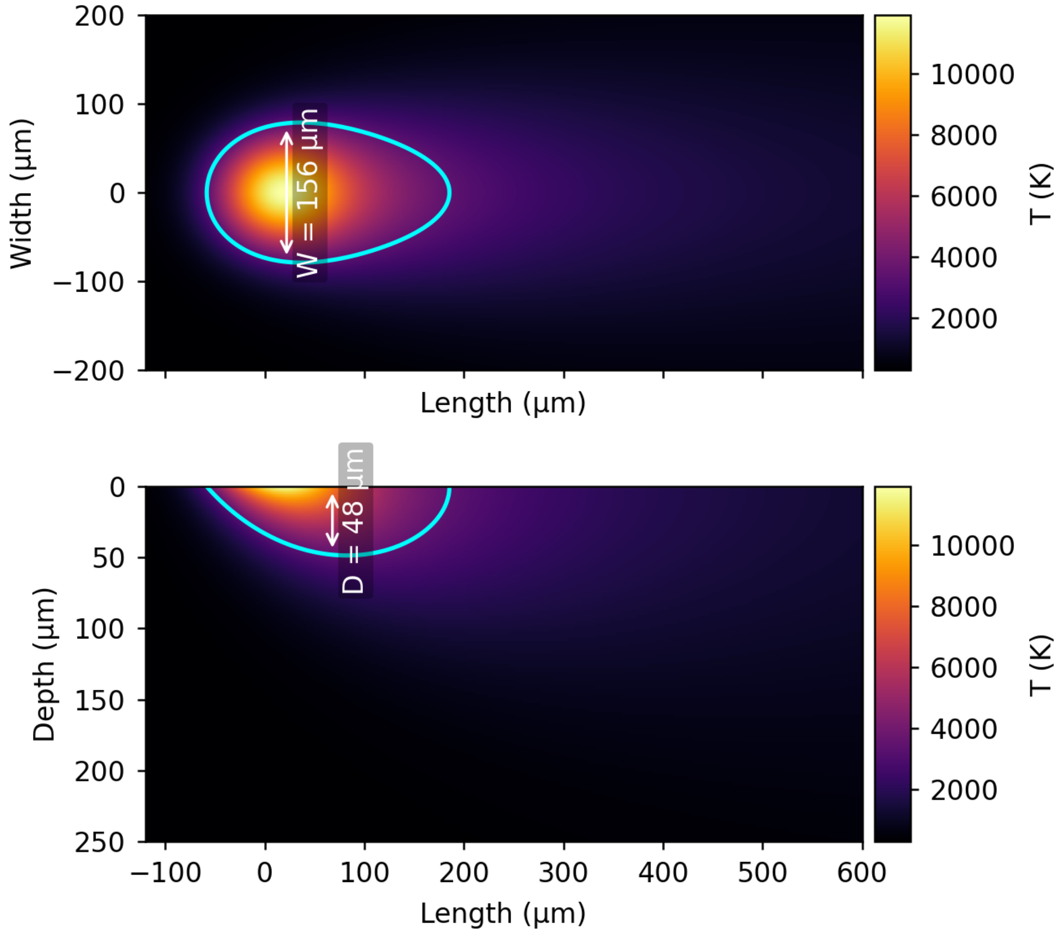
Refactored Eagar–Tsai Model

Python library implementation with compiled-C speedup

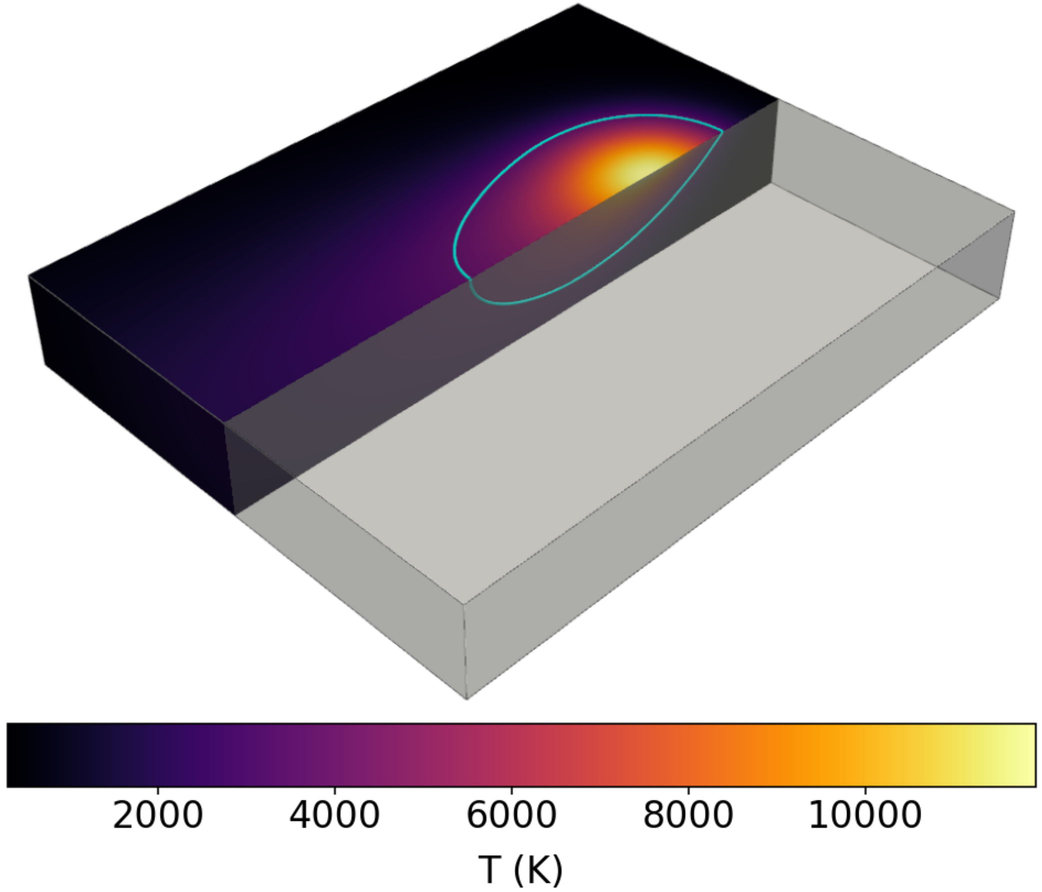


- Separates melt-pool prediction, printability maps, 2D heatmap generation, and 3D temperature-field export into reusable steps.
- Compiled C integrand removes Python overhead inside QUADPACK for faster parameter sweeps.

ET Thermal Cross Sections

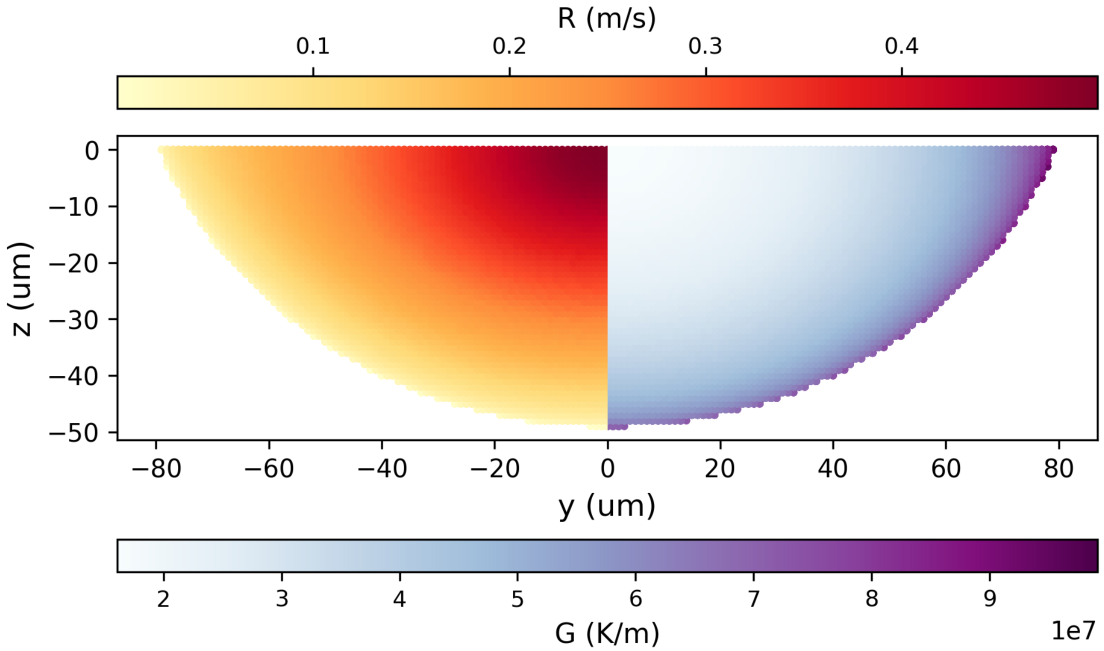


Temperature heatmap for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.



3D view of the ET melt pool for the same alloy and process condition.

ET Thermal Gradient and Solidification Rate



Projection of G and R along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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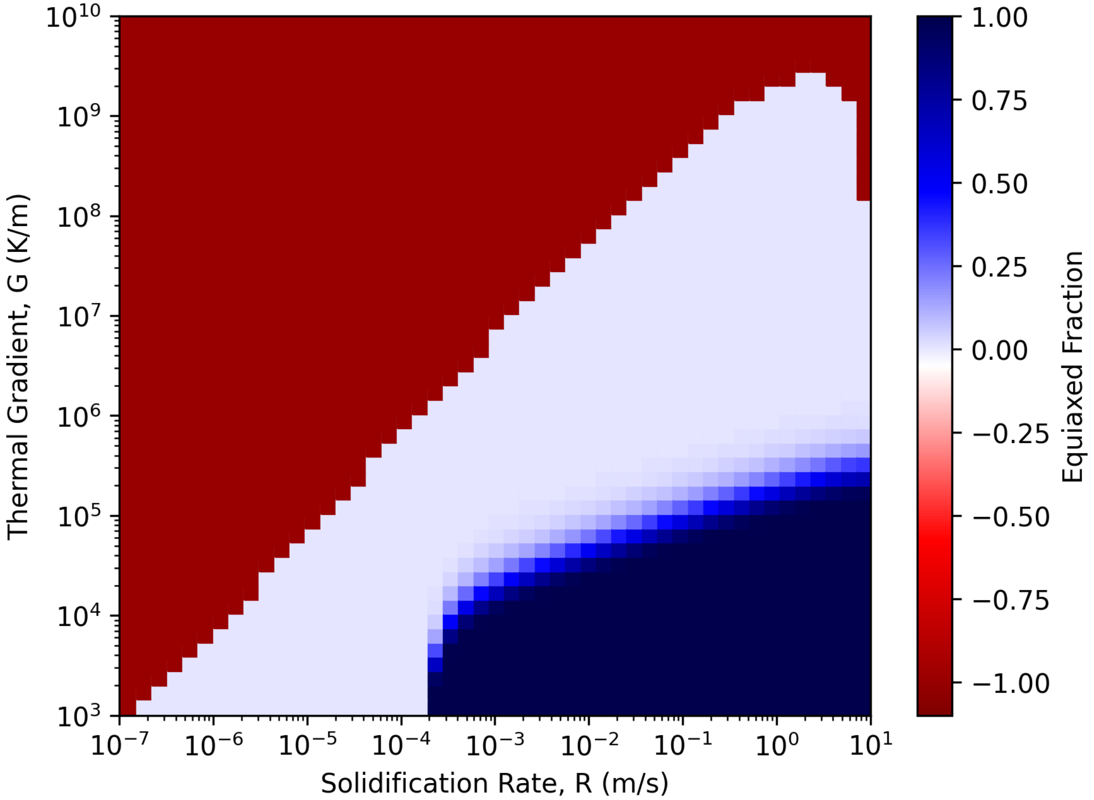
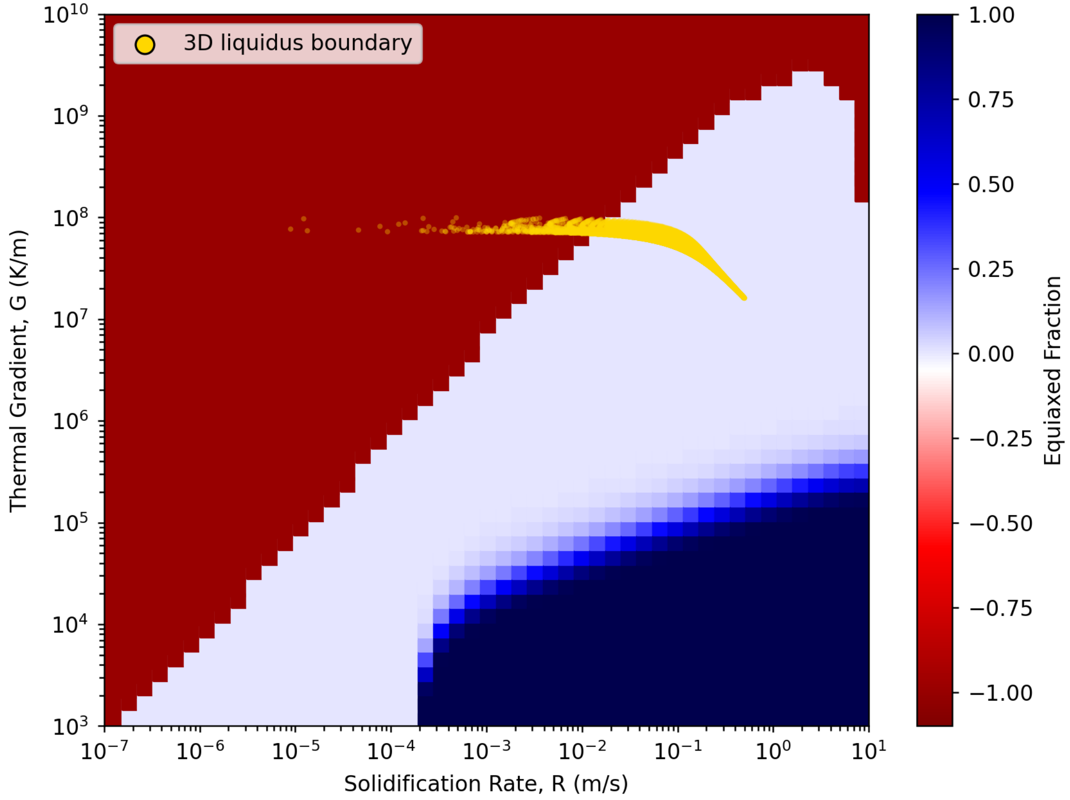


GR Map calculated with Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

ET Microstructure Classification



GR Map calculated with Thermo-Calc CET Property Model with overlaid GR liquidus points for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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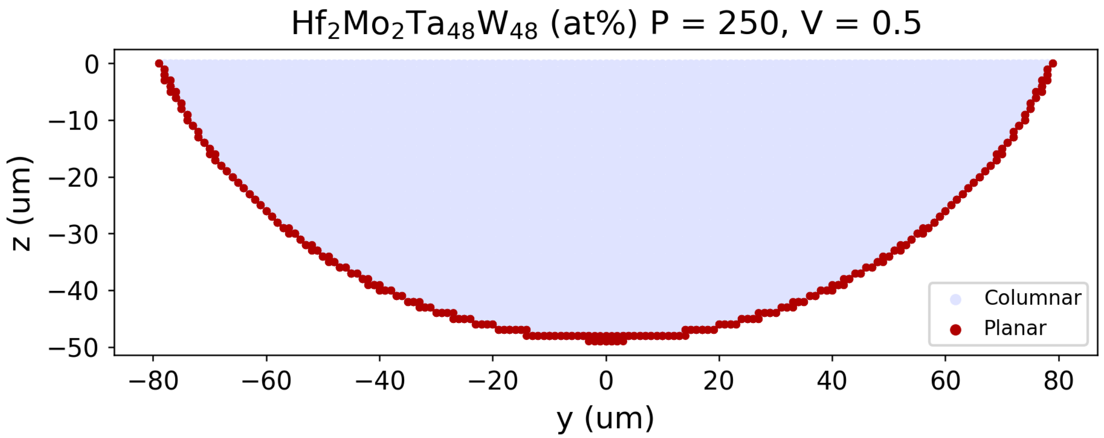
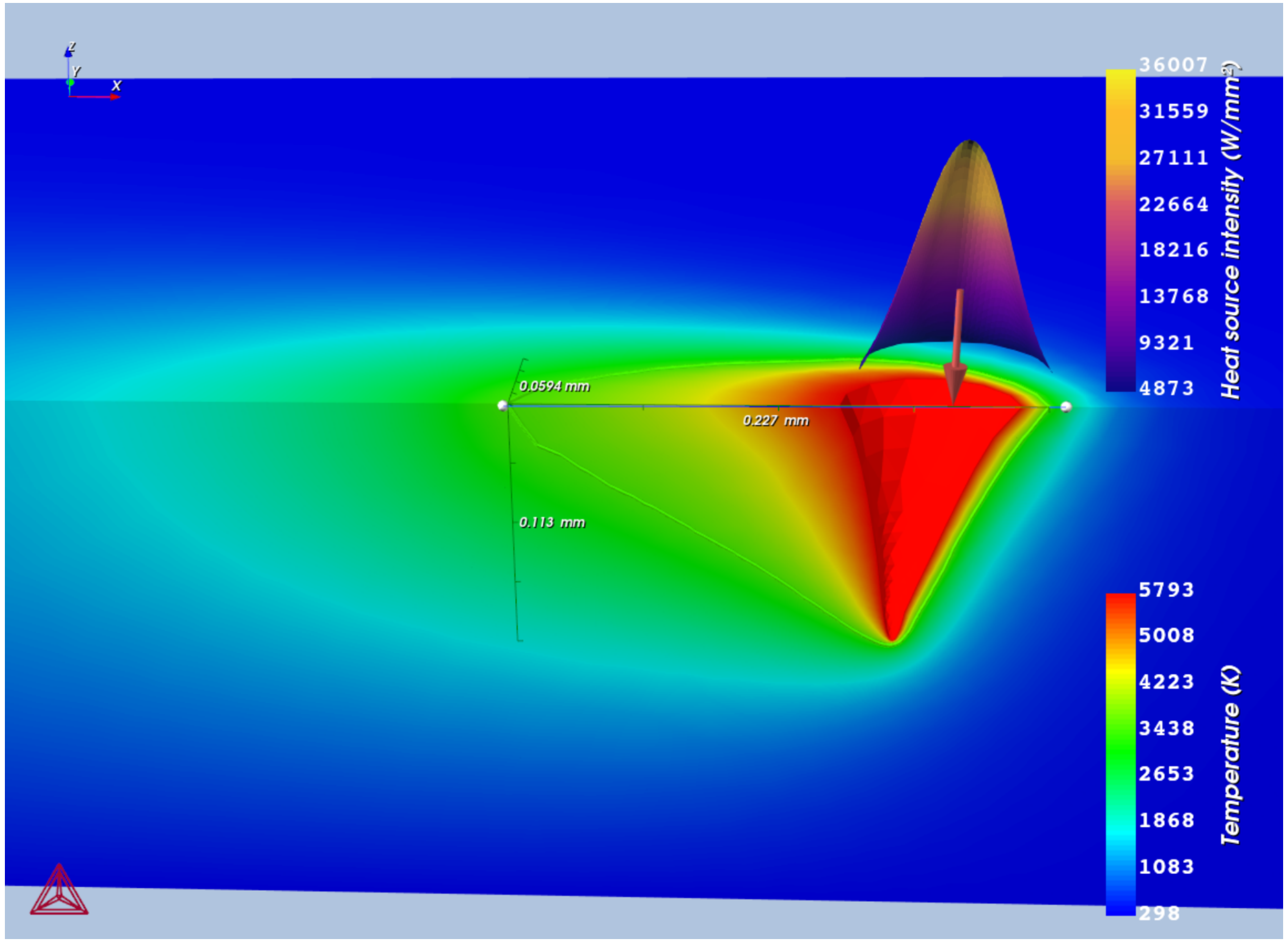


Projected microstructure classification along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

Thermo-Calc AM Module



Steady-state TCAM simulation for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

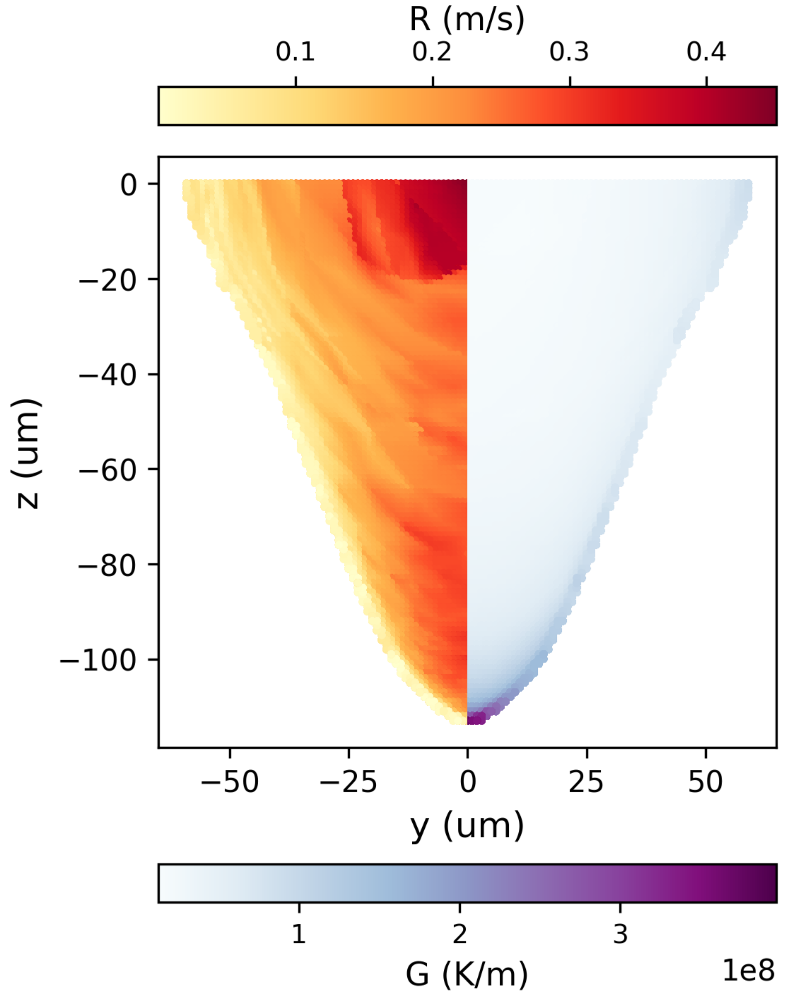
Higher-Fidelity Thermal Reference

- Integrates FEM for thermal conduction with CFD for melt-pool fluid flow.
- Composition-specific, temperature-dependent thermophysical properties.
- Captures Marangoni flow, keyholing, evaporation, radiation, and convection.

Outputs Used in This Work

- 3D temperature field
- Melt-pool geometry
- Thermal gradient (G) and solidification rate (R) from ParaView post-processing

TCAM Thermal Gradient and Solidification Rate



TCAM projection of G and R along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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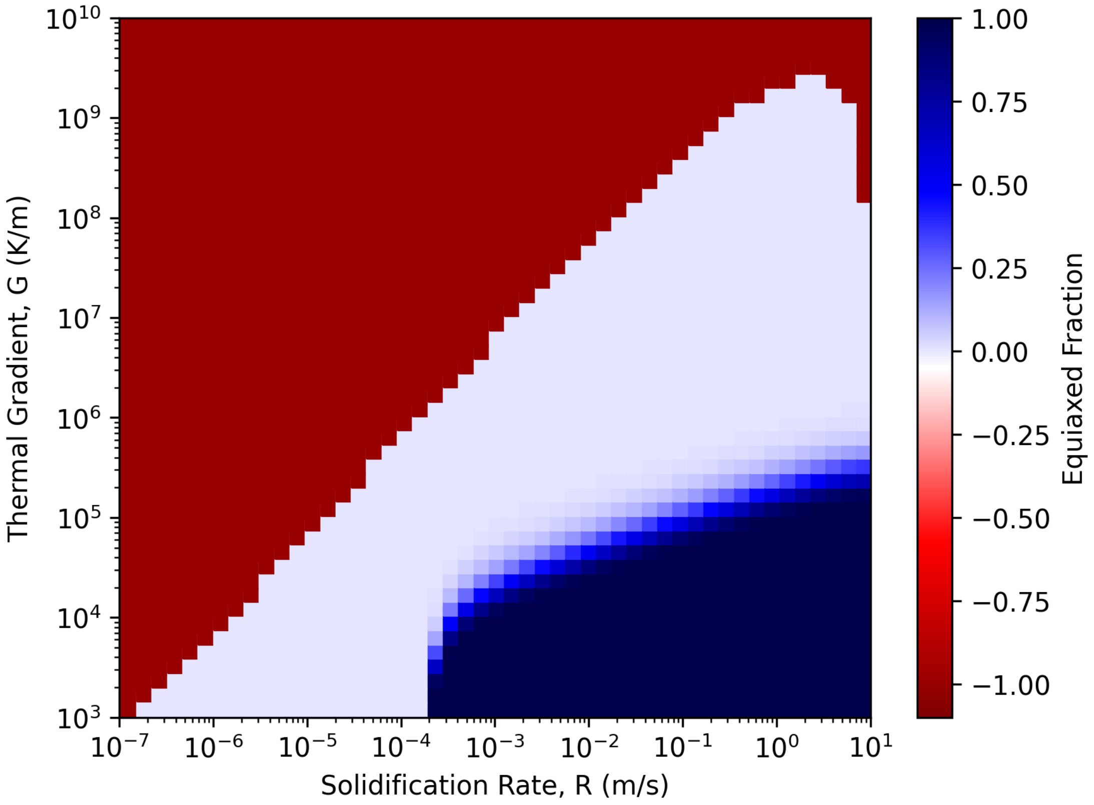
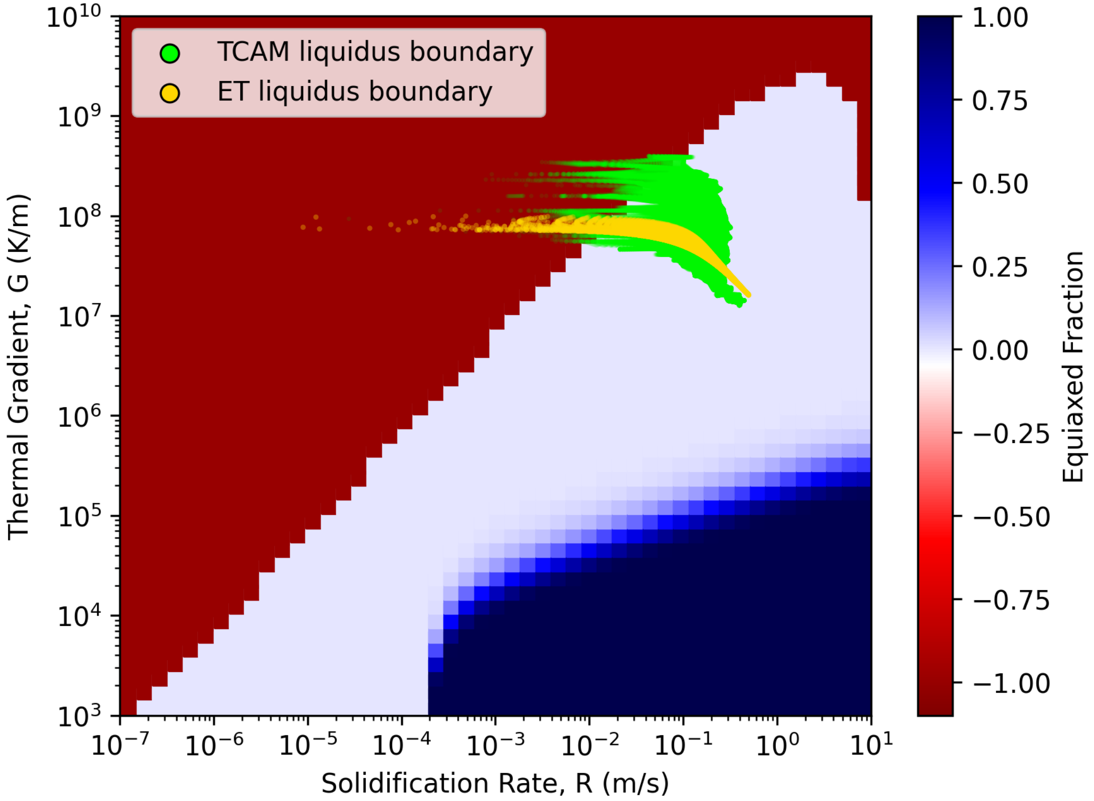


GR Map calculated with the Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

TCAM Microstructure Classification



GR map with TCAM and Eagar–Tsai liquidus-boundary conditions overlaid for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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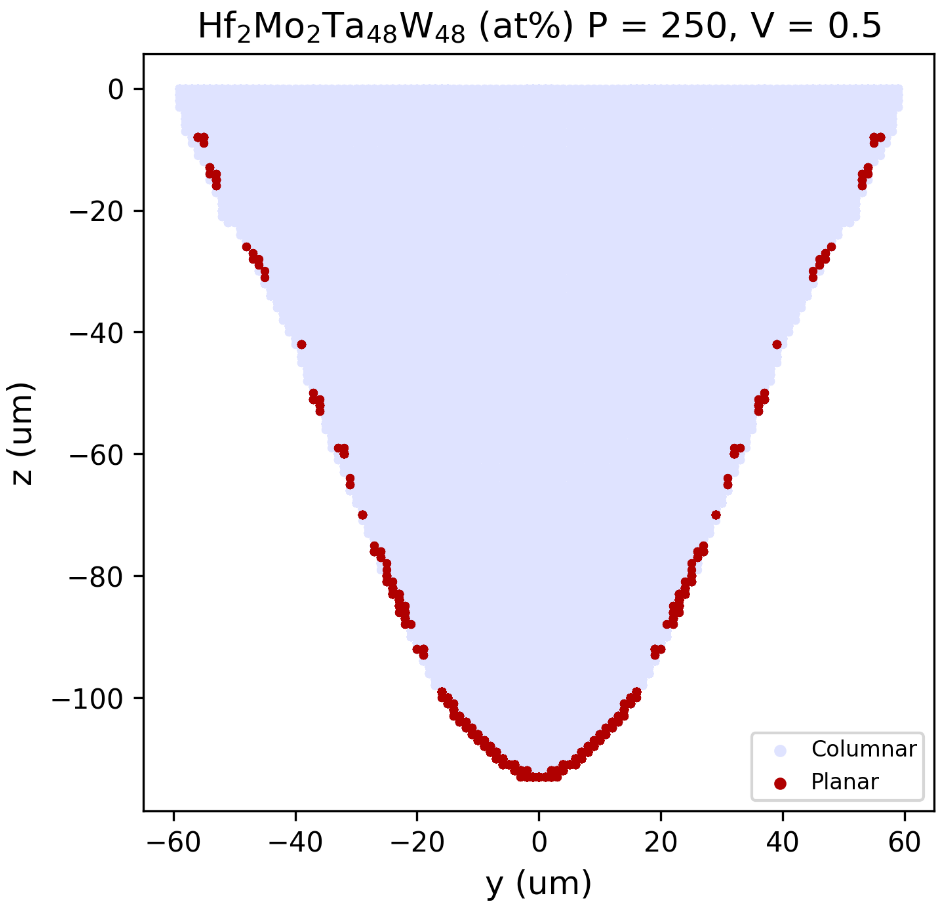
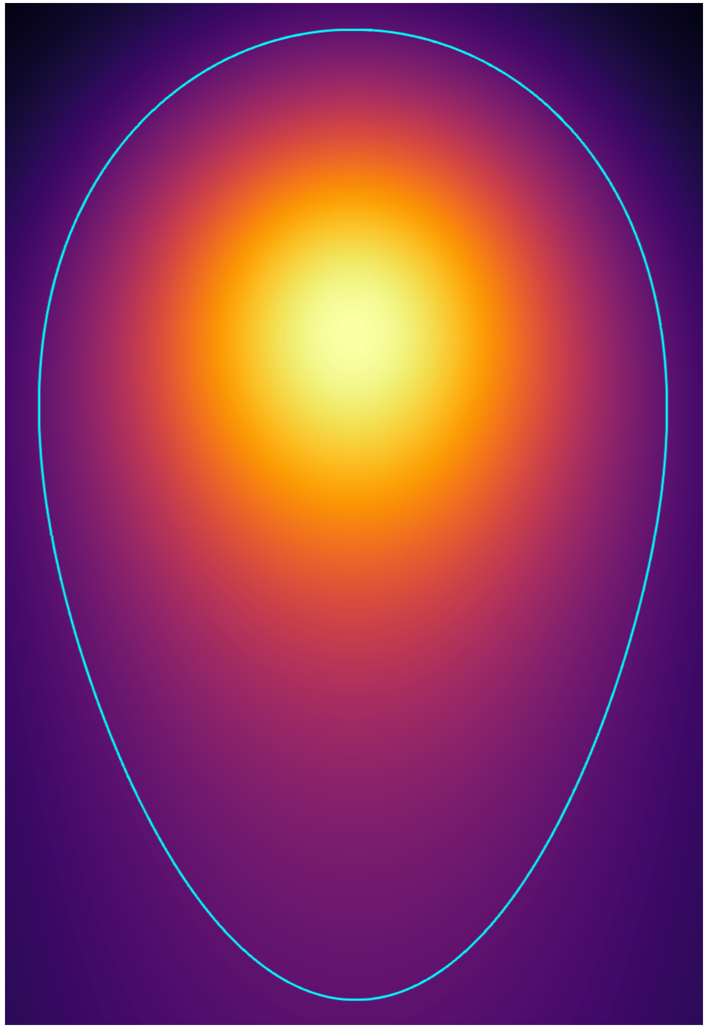
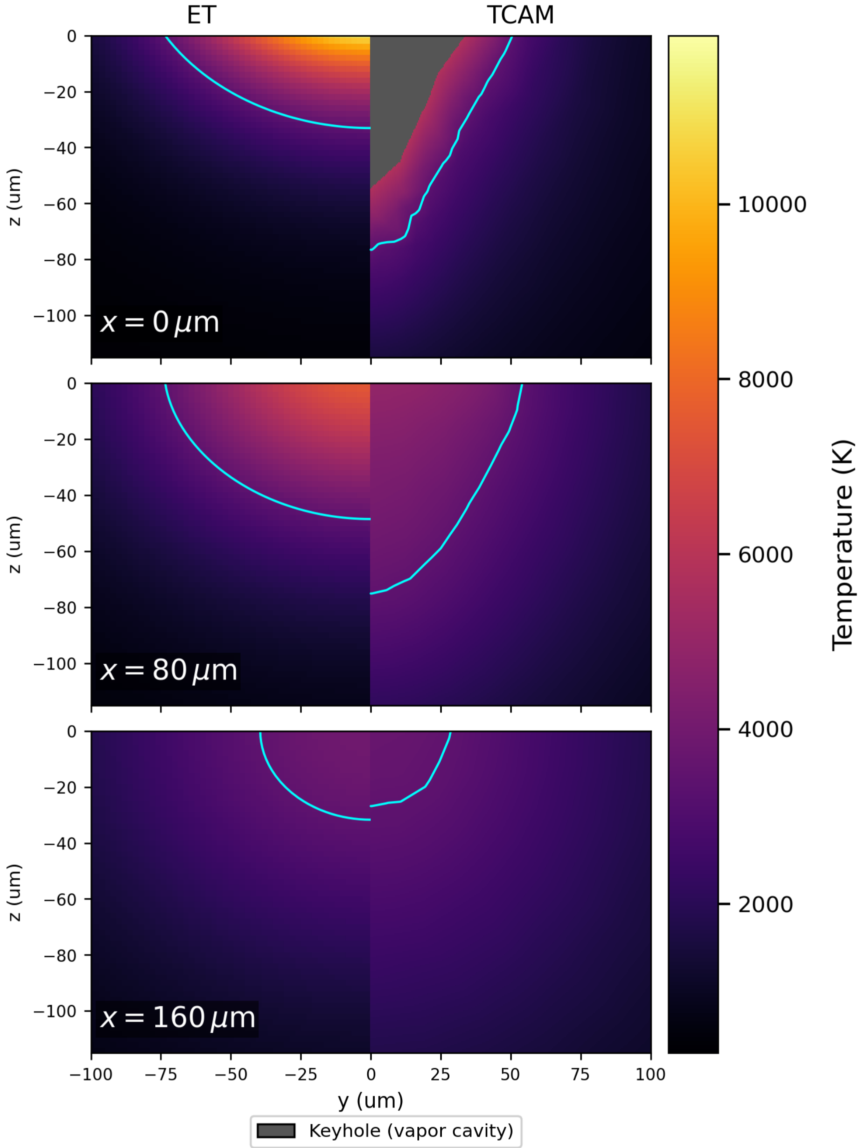
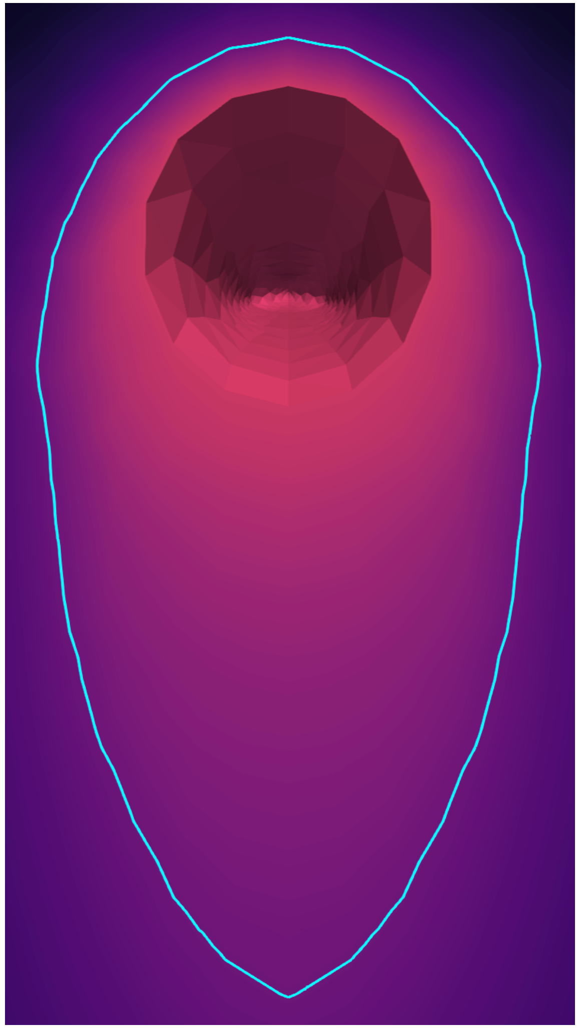


TCAM-derived microstructure classification projected along the solidification front onto a yz slice for the same alloy and process condition.

ET vs. TCAM – Melt Pool Comparison







Closing

Takeaway

End with the decision, ask, or technical conclusion you want the audience to remember.

- **Deliverable**: Name the concrete output.
- **Primary risk**: Name the hardest unresolved issue.
- **Immediate next step**: Name the action that starts now.